

5(a)(i)	<i>any two from: (transgenic pigs – ora for normal pigs)</i> 1 GH (concentration) higher / GH always present not just at certain times ; 2 hGH, works for longer / broken down less quickly (than pig GH) ; 3 increase in cell signalling ; 4 increased / activates, (growth) gene, expression / transcription ; 5 increased / stimulates, cell division / mitosis ; 6 (make) more, muscle / bone / fat ;	2
5(a)(ii)	<i>any three from:</i> 1 (mouse DNA / it) is, the <u>promoter</u> ; 2 (where) RNA polymerase / transcription factor(s), bind ; 3 controls / allows / ensures / is needed for, gene expression / gene activation / mRNA production / hGH production ; 4 gene can be, switched on / transcribed, by adding metal ions ; 5 <i>idea of</i> controlling, when / where / how much, the gene is, expressed / transcribed ;	3
5(b)(i)	<i>NO arguments (any two to question max three):</i> 1 few offspring / don't pass on gene(s) / unsustainable / assisted reproduction is expensive (as problem with mating) ; 2 cost of healthcare / less growth / die young (due to ulcers) ; 3 people may, avoid / refuse to buy / pay less for, GM food ; 4 (GM) production cost, expensive / outweighs benefits (as 1% success) ; <i>YES arguments (any two to question max three):</i> 1 greater yield / more meat (as higher, body mass / muscle) ; 2 higher price / worth more money, as more muscle to fat ; 3 sell / slaughter / process, earlier / at younger age ;	3

Question	Answer	Marks
5(b)(ii)	<i>any two from:</i> <i>unethical / not good / not right, because:</i> 1 pigs, suffer / have (stomach) ulcers / experience pain ; 2 pigs cannot, behave normally / move much / exercise / keep fit / socialise / mate ; <i>ethical / good / right, because:</i> 1 more / better quality, meat / food, for humans ; 2 pigs suffer less than (normal) pigs given hGH by injection ;	2
5(c)(i)	<i>any two from:</i> success rate (in altering gene) is, greater / 100% (instead of 1%) ; (only) specific gene is altered / targets gene more precisely ; (unwanted) gene is, removed / deleted / disabled / knocked out ;	2
5(c)(ii)	<i>one or two arguments about statement's truth to question max 3:</i> 1 not selective breeding as, DNA / genes, manipulated / altered / changed / removed ; 2 not selective breeding as IVF is used ; 3 not selective breeding as no, crossing and, selection (of offspring) / repetition ; 4 <u>not</u> genetic engineering as no new gene is put into, organism / zygote ; <i>one or two reasons why more acceptable to question max 3:</i> 5 no foreign gene inserted / no cross-species gene transfer ; 6 (only) one / single, gene, altered / silenced ; 7 (application) reduces, suffering of / (viral) disease in, pigs ;	3